ASSER ISMAIL

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Toronto, ON (Open to Relocation Anywhere) (Eligible to work in the US; no sponsorship required)

PROFILE

- Software Engineer experienced in AI/ML & Distributed Systems, building end-to-end production MCP and RAG systems
- Co-inventor on a 2025 Patent in RDMA-based replication for distributed cloud databases
- Honours B.Sc. in Computer Science from the University of Toronto & Member of the Dean's List Scholar

TECHNICAL SKILLS

Languages: Python, Java, C, C++, TypeScript, JavaScript, HTML, CSS, SQL, Shell, Assembly
Software: AWS, Docker, Kubernetes, Firebase, Flask, FastAPI, LangChain, LangGraph, OpenAI API, Gemini API, Ollama, REST APIs
Tools: React.js, Databricks, Git, Linux, Redis, Postman, PyTorch, TensorFlow, Jupyter Notebook, GitLab, GitHub, Claude Code, Codex

EXPERIENCE

AI Software Engineer | AI Samurai *Remote - US* ***Oct. 2025 - Oct. 2026***

- Developed AI-powered media summarization pipelines using OpenAI and Gemini APIs for 100+ audio and video content
- Integrated LangGraph based agentic workflows for multi-step reasoning and tool calling, and structured summary generation
- Improved AI summary quality by 25% through prompt optimization, benchmarked using DeepEval against baseline outputs
- Built backend processing workflows integrating LLM APIs with media ingestion and summarization services

Cloud Database DevOps Engineer | Huawei Canada *Markham, ON, CAN* ***May 2024 - Sep. 2025***

- Developed low-level C, Python, and Shell software for Linux cloud database systems, supporting enterprise SaaS and 5G
- Improved deployment efficiency by 40% using Python/Shell automation, benchmarked through GitLab CI/CD execution logs
- Debugged MySQL and PostgreSQL database systems, investigating replication, failover, and performance regressions
- Designed synthetic data generation pipelines using LLMs from DDL schemas for database testing and benchmarking
- Improved semantic retrieval speed by 23% using Ollama, Vanna AI, and vector databases, validated by pytest benchmark
- Co-inventor on a patent (2025) for DMA pull-based RDMA log sync. between master/replica in cloud-native databases

PROJECTS

Multi-Agent Customer Support AI | ***Python, LangGraph, MCP, RAG, FastAPI, Oracle SQL, Chroma***

- Built a LangGraph multi-agent system with supervisor routing, SQL/RAG agents, and context-aware response synthesis
- Developed an MCP server with Oracle schema discovery, metadata indexing, and table relationship mapping
- Implemented secure read-only SQL execution with query validation, access controls, and automated testing
- Built a RAG pipeline using OpenAI embeddings and Chroma for PDF ingestion, semantic search, and source citations

CNN Photo Geolocation Model | ***Python, PyTorch, EVA-02, CNN, Google Street View API***

- Designed a CNN-based photo geolocation model using continent classification and coordinate regression on Street View image
- Built a 38K+ image dataset from the Google Street View API with multi-angle sampling and data augmentation for robustness
- Applied transfer learning with the EVA-02 backbone pretrained on ImageNet-22k to extract geographic visual features
- Achieved 96% continent accuracy and a 251 km avg. localization error, outperforming Google PlaNet and Claude-3.5 baselines

Distributed Real-Time Pixel Platform | ***Python, AWS Lambda, API Gateway, WebSockets, Redis, CloudFront***

- Built a serverless real-time pixel platform using AWS Lambda, API Gateway WebSockets, Redis and CloudFront
- Managed WebSocket connections and distributed rate limiting with Redis to support concurrent users
- Optimized performance with atomic Redis operations and CloudFront edge caching to reduce latency

EDUCATION

University of Toronto| Computer Science Specialist
Honours Bachelor of Science (HBSc) + PEY Co-op

Relevant Coursework:
- Machine Learning & AI (PyTorch, Python)
- Deep Learning (PyTorch, TensorFlow, Architecture)
- Computer Networks (TCP/IP, Routing, Network Protocols)
- Software Tools and System Programming (C, Shell)
- Operating Systems (C, Linux, Processes, Threads)
- Database Systems (SQL, PostgreSQL, MySQL)

CERTIFICATES & HONORS

2026 **Agentic AI,** Built autonomous AI agents using planning, reflection, tool use, and multi-agent workflows **DeepLearning.AI**
2026 **Agile Project Management,** Learned Scrum, Kanban, backlog, sprint planning and Agile delivery practices **Google**
2026 **InstructLab Bias Detection Model,** Fine-tuned Granite-7B using synthetic data to diagnose bias **Riipen/Red Hat x IBM**
2024 **ML Specialization**, Completed supervised, unsupervised, & reinforcement learning courses (NLP, Transformers) **Coursera**
2023 **Microsoft Certified: Azure Fundamentals,** Cloud concepts in networking and Azure in particular **Microsoft**
2023 **Dean's List Scholar,** Maintained 3.50+ CGPA throughout the academic year **University of Toronto**